WAB

Provadis School of International Management and Technology

Analyzing the Correlation Between Cyclomatic Complexity and Code Quality in Python Projects

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Abstract

Abstrakt

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AI Declaration

The usage of AI tools within this project is documented here. I declare that I have documented all interactions with AI tools, including the prompts used and the outputs received.

System	Prompt	Usage
GitHub Copilot 1	I want you to rewrite my template from LaTeX to Typst, and make it look like it did before.	Recreated the template in Typst, maintaining the original appearance.
GitHub Copilot 2	In Code/, write a Python script that finds the top most depended-upon Python packages and analyzes the correlation between function cyclomatic complexity and bug-fix frequency over an adjustable time window, with plots and tables.	Created the analysis pipeline.
GitHub Copilot 3	Use pip to install any required packages, not conda.	Added pip dependency setup.
GitHub Copilot 4	Correct the package discovery so the ranking reflects dependency counts and the analysis uses real source repositories rather than release repositories.	Fixed discovery and repo resolution.
GitHub Copilot 5	Make the analysis practical to run by adding adjustable bounds for the mined recent commit history.	Added bounded commit mining.
GitHub Copilot 6	Use ProcessPoolExecutor to accelerate the script.	Parallelized package mining.
GitHub Copilot 7	Add compact logging and terminal progress bars, show phases such as cloning, mining, SZZ, and plotting, and remove Typst table generation.	Added logs and progress bars; removed Typst exports.
GitHub Copilot 8	Fix the parser warning noise produced while mining repositories.	Suppressed parser warnings.

System	Prompt	Usage
GitHub Copilot 9	Fix the Windows UnicodeDecodeError that occurs while reading subprocess output during mining.	Hardened subprocess decoding.
GitHub Copilot 10	Implement flexible caching so repeated runs can reuse already mined information while still respecting parameter changes.	Added parameter-aware mining cache.
GitHub Copilot 11	Add analysis support for native code too and update the progress bars more often.	Added mixed-language analysis; increased progress updates.
GitHub Copilot 12	Use all languages recognized by lizard instead of restricting the non-Python analysis to C-family files.	Expanded lizard language coverage.

Declaration of Authorship

I hereby confirm that I have personally and independently prepared the present work and have not used any sources or aids other than those specified. All passages taken verbatim or in substance from other sources are identified as such. The drawings, illustrations and tables in this work are created by me or have been provided with an appropriate source reference. This work has not been submitted by me to any other university in the same or similar form for the acquisition of an academic degree.

Frankfurt, 03.05.2026

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